

FTS Midterm Exam IV (Weeks 2–5)

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Financial Time Series Analysis · Midterm Examination · Prof. Keehong Rhee

Scope: Weeks 2–5 — stationarity, AR and MA (including how to rewrite one as the other), ARIMA and seasonality, volatility. 40 multiple-choice questions, 2.5 points each, 100 points total. The questions test ideas, not memorised numbers. Several follow Mina, a new analyst, as she learns to read time series.

Item	Detail
Name	
Student ID	
Time allowed	75 minutes
Materials	Closed book; a non-programmable calculator is allowed

Instructions. Choose the ONE best answer (A–D). There is no penalty for wrong answers. "Stationary" means weakly stationary: the average, the spread and the link between today and k days ago do not change over time.

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1		11		21		31	
2		12		22		32	
3		13		23		33	
4		14		24		34	
5		15		25		35	
6		16		26		36	
7		17		27		37	
8		18		28		38	
9		19		29		39	
10		20		30		40	

Section I · Does the Series Stay Put? (Q1–12)

Q1. Mina plots a stock over ten years. The price drifts upward and never settles around a fixed level. The daily returns bounce around zero with a similar size of swings throughout. Which series is most likely stationary?

- A) The price
- B) The daily returns
- C) Both the price and the returns

- D) Neither of them

Q2. Which description best matches a weakly stationary series?

- A) The series is a flat line that never moves.
- B) The series rises at a constant speed.
- C) The series always follows a normal (bell-shaped) distribution.
- D) Its average, its spread, and the link between today and k days ago stay the same over time.

Q3. A series keeps a steady average, but its swings get bigger every year. Is it weakly stationary?

- A) No — its spread (variance) is not constant over time.
- B) Yes — the average is constant, and that is all stationarity needs.
- C) Yes — as long as the values look bell-shaped.
- D) It cannot be judged without a forecast.

Q4. Think of a dog tied to a post. Two series follow the model below, one with $\phi = 0.5$ and one with $\phi = 1$.

$$X_t = \phi X_{t-1} + \varepsilon_t$$

Which statement is correct?

- A) Both series are pulled back to their average.
- B) Both series are random walks.
- C) With $\phi = 0.5$ the series is pulled back toward its average (a dog on a leash); with $\phi = 1$ it is a random walk whose shocks never fade (no leash).
- D) With $\phi = 1$ the series returns to its average faster, because all of yesterday carries over.

Q5. Mina simulates 25 random walks that all start at zero. After 100 days the paths are spread much wider than after 10 days. Why?

- A) The average of a random walk drifts upward over time.
- B) Random walks always have fat tails.
- C) Every step adds a new shock and no shock ever fades, so the uncertainty keeps piling up (the variance grows with time).
- D) The simulation must contain an error.

Q6. What happens when you take the first difference (today minus yesterday) of a random walk?

- A) You get back the shocks themselves — white noise, which is stationary.
- B) You create a new upward trend.
- C) The series becomes explosive.
- D) Nothing changes; it is still a random walk.

Q7. The ACF of a price series starts near 1 and decays very slowly. The ACF of its daily returns drops to almost zero right after lag 0. What does this tell Mina?

- A) The price is stationary, but the returns are not.
- B) Both series have unit roots.
- C) A slowly decaying ACF means a sure trading profit.
- D) The price behaves like a unit-root series and should be differenced; the returns already look stationary.

Q8. After the same large shock, series A soon returns to its old straight trend line, while series B stays at its new lower level for good. How should each be handled?

- A) Difference both series.
- B) A is trend-stationary, so remove the trend line; B has a unit root, so difference it.
- C) A has a unit root; B is trend-stationary.
- D) Remove the trend line from both series.

Q9. Mina differences a series that was already stationary. The new series shows a large negative correlation between neighbouring days. What happened?

- A) She differenced one time too many (over-differencing).
- B) She discovered strong momentum in the data.
- C) The series turned into a random walk.
- D) The data must be wrong.

Q10. A walker and her dog on a leash both wander around the city, yet the distance between them always stays within a few metres. In the same way, two stock prices each wander, but their gap $Y - \beta X$ keeps returning to a fixed level. What are the two prices called?

- A) Highly correlated returns
- B) Cointegrated
- C) Two trend-stationary series
- D) Two white-noise series

Q11. Stocks Y and X are cointegrated with spread $Y - 1.5X$. The spread normally sits around 10 with a standard deviation of 2. Today $Y = 106$ and $X = 60$, so the spread is 16. What does a simple pairs trade do?

- A) Buy Y and sell X, because Y has been rising.
- B) Buy both stocks, because they move together.
- C) Do nothing, because the spread is within its normal range.
- D) The spread is 3 standard deviations above normal ($z = +3$), so Y is rich relative to X: sell Y, buy 1.5 units of X, and close the trade as the spread returns toward 10.

Q12. Mina sold Y and bought 1.5 units of X when the spread's z-score reached +3. A week later one of the two companies announces a merger. The spread keeps widening ($z = +5$), and a fresh check shows the two prices are no longer cointegrated. What should she do?

- A) Add to the trade — a bigger z-score means a stronger signal.
- B) Hold the trade forever, because spreads always come back.
- C) Switch to buying Y, because it keeps going up.
- D) Treat it as a broken leash: the long-run tie between the prices is gone, so close the trade (stop-loss) instead of waiting for the spread to return.

Section II · Memory: AR, MA and Switching Between Them (Q13–26)

Q13. One unit shock hits a stationary series. Its effect today, tomorrow, the day after, and so on is 1, 0.5, 0.25, 0.125, ... — like a ball that keeps half its height on every bounce. Which model produces this pattern?

- A) An AR(1) with $\phi = 0.5$: the echo shrinks every day but never fully disappears.
- B) An MA(1) with $\theta = 0.5$
- C) White noise
- D) A random walk

Q14. A news surprise moves a series today and still has a smaller effect tomorrow. From the day after tomorrow on, its effect is exactly zero — like a rumour that is forgotten after one day. Which model fits?

- A) An AR(1) with $\phi = 0.5$
- B) A random walk
- C) An MA(1)
- D) An AR(2)

Q15. Think of the telephone game: Grandma (day $t-2$) tells Mom (day $t-1$), and Mom tells the Child (day t). Which statement correctly describes the ACF and the PACF at lag 2?

- A) The ACF counts only the direct line from Grandma to the Child; the PACF counts every path.
- B) The ACF counts every path, including the one through Mom; the PACF counts only a direct line after Mom's part is removed.
- C) The ACF and the PACF always measure the same thing.
- D) The PACF can only be used for price levels.

Q16. For a stationary series, the ACF is clearly non-zero at lags 1 and 2 and near zero after that. The PACF fades away gradually. Which model does this suggest?

- A) AR(2)

- B) White noise
- C) ARMA(2,2)
- D) MA(2)

Q17. For another stationary series, the PACF has one clear spike at lag 1 and is near zero after that. The ACF fades away gradually (0.7, 0.49, 0.34, ...). Which model does this suggest?

- A) MA(1)
- B) AR(1)
- C) White noise
- D) A random walk

Q18. Both the ACF and the PACF of a stationary series fade away gradually; neither one stops suddenly. What is the most sensible first model to try?

- A) A small ARMA model, such as ARMA(1,1), that mixes both kinds of memory
- B) A pure MA model with a very long memory
- C) A random walk
- D) No model at all, because the series cannot be described

Q19. Mina estimates an AR(1) and finds $\phi = 1.2$. What is wrong?

- A) Nothing — a larger ϕ means the series is more stationary.
- B) The model has become a moving average.
- C) The series would explode: an AR(1) is stationary only when $|\phi| < 1$.
- D) The model has become white noise.

Q20. (AR $\rightarrow$ MA) A stationary AR(1) with $\phi = 0.8$ is rewritten using only today's and past shocks. Substituting yesterday's value again and again gives the form below. What does the result look like?

$$X_t = \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-2} + \phi^3 \varepsilon_{t-3} + \dots$$

- A) An MA(1) with $\theta = 0.8$
- B) It cannot be rewritten, because an AR depends on past values, not past shocks.
- C) An MA with endless terms whose weights shrink: 1, 0.8, 0.64, 0.512, ... — a stationary AR(1) is an MA(∞).
- D) An MA with endless terms that all have the same weight 0.8

Q21. (Why AR $\rightarrow$ MA?) Mina can write her AR(1) with $\phi = 0.8$ as an MA(∞), as in Q20. Why is this rewrite useful?

- A) It reduces the number of parameters to estimate.
- B) AR models cannot be estimated from data, so the MA form is needed.

- C) It turns a non-stationary series into a stationary one.
- D) It shows directly how much of each past shock is still inside today's value — how fast a shock dies out — and the weights must shrink for the series to be stationary.

Q22. (MA → AR) Now go the other way. Start from an MA(1) with $\theta = 0.5$:

$$X_t = \varepsilon_t + 0.5 \varepsilon_{t-1} \Rightarrow \varepsilon_t = X_t - 0.5 \varepsilon_{t-1}$$

Replacing yesterday's shock again and again by past values of X , what form do you reach?

- A) $X_t = 0.5X_{t-1} + \varepsilon_t$, a plain AR(1)
- B) An AR with endless terms, $X_t = 0.5X_{t-1} - 0.25X_{t-2} + 0.125X_{t-3} - \dots + \varepsilon_t$: weights that alternate in sign and shrink — an MA(1) is an AR(∞).
- C) It cannot be done, because an MA has no past values of X .
- D) An AR with endless terms whose weights grow: 0.5, 1, 2, 4, ...

Q23. (Why MA → AR?) In an MA(1), today's value is built from today's and yesterday's shocks, but nobody ever observes the shocks — Mina only sees the data X . Why do we rewrite an MA(1) as an AR(∞)?

- A) To express the model in past values of X that we can actually observe, so the hidden shocks can be recovered and forecasts made from the data — which works only when $|\theta| < 1$ (invertible).
- B) To make the series stationary.
- C) Because an MA model has no memory at all.
- D) To add more parameters and improve the fit.

Q24. (The big picture) Q20–Q23 show that a stationary AR can be written as an MA(∞), and an invertible MA as an AR(∞). What is the main lesson?

- A) AR models are always better than MA models.
- B) MA models only work for non-stationary data.
- C) AR and MA are two ways of describing the same memory; choose the form that needs fewer parameters (a long AR may be a short MA, and vice versa).
- D) Rewriting the model changes the data it describes.

Q25. An interest-rate spread follows a stationary AR(1) with an average of 2%. Today it is 4%. How do its forecasts behave as Mina looks further ahead?

- A) The forecast stays at 4% at every horizon.
- B) Like a stretched rubber band, the forecast moves back toward 2%; far-ahead forecasts are close to 2%, with a wider uncertainty band.
- C) The forecast jumps straight to 2% tomorrow.
- D) The forecast keeps rising above 4%.

Q26. After fitting a model, Mina plots the ACF of what is left over (the residuals). One clear spike remains at lag 1. What should she conclude?

- A) The model has missed some memory; she should revise it (for example, add a term) and check again.
- B) The model is perfect.
- C) Returns must be predictable.
- D) She should stop and use the model as it is.

Section III · Trends and Seasons: ARIMA and SARIMA (Q27–32)

Q27. Mina's sales series climbs like a staircase, so an ARMA model does not fit it directly. What does the "I" step in ARIMA do?

- A) It adds an extra seasonal term.
- B) It removes the biggest outliers.
- C) It forces the forecast back to the average.
- D) It differences the series to remove the trend, fits an ARMA to the changes, and then adds the changes back up to forecast the level.

Q28. The first difference of a price series looks like pure white noise. Which ARIMA model describes the price itself?

- A) ARIMA(1,0,0)
- B) ARIMA(0,1,0) — the random walk
- C) ARIMA(0,0,1)
- D) ARIMA(0,2,0)

Q29. Mina compares two forecasts made today. Model P is a stationary AR(1) for an interest-rate spread. Model Q is an ARIMA(0,1,0) with drift for a stock price. How do their long-range forecasts differ?

- A) P returns to its long-run average; Q continues from the latest level along the drift, with an uncertainty band that keeps widening.
- B) Both return to their long-run averages.
- C) Both stay exactly at today's value.
- D) P keeps trending, while Q returns to an average.

Q30. After one difference, Mina's series looks stationary. After a second difference, it shows a large negative correlation between neighbouring days and bigger swings. How many differences (d) should she use?

- A) $d = 0$

- B) $d = 2$
- C) $d = 1$ — the smallest number that makes the series stationary; the second difference was one too many
- D) $d = 3$

Q31. Monthly ice-cream sales jump every summer. The ACF of the sales shows spikes at lags 12, 24 and 36. What does this pattern show?

- A) A random walk
- B) Over-differencing
- C) White noise
- D) A yearly seasonal pattern with period $s = 12$

Q32. To remove that yearly pattern, Mina compares each month with the same month one year earlier (this June – last June). What is this step called, and what does it do?

- A) A seasonal difference with $s = 12$: it removes the pattern that repeats every year.
- B) A regular first difference: it removes the seasonal spikes at lags 12 and 24.
- C) Detrending: it removes a straight trend line.
- D) A moving average: it adds one month of memory.

Section IV • Forecasting the Storm: ARCH, GARCH and VaR (Q33–40)

Q33. Mina checks daily stock returns. The ACF of the returns is close to zero at every lag, but the ACF of the squared returns is clearly positive for many lags. What has she found?

- A) Returns are easy to forecast in direction.
- B) The returns must contain a unit root.
- C) The direction of returns is unpredictable, but their size has memory: big days follow big days (volatility clustering).
- D) The data must be wrong, because the two ACFs disagree.

Q34. What is the difference between conditional and unconditional variance?

- A) They are two names for the same number.
- B) Conditional variance is the long-run average; unconditional variance changes every day.
- C) Conditional variance applies only to prices, unconditional variance only to returns.
- D) Unconditional variance is one long-run number; conditional variance is today's variance given recent days, so it rises after big shocks and falls in calm periods.

Q35. In an ARCH(1) model, today's variance depends on yesterday's squared shock:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2$$

Yesterday the market fell 5%. On another day it rose 5%. How does each move affect today's variance?

- A) The fall raises today's variance; the rise lowers it.
- B) Both raise today's variance by the same amount, because the shock is squared and only its size matters.
- C) Neither has any effect.
- D) Only the rise matters.

Q36. GARCH(1,1) builds today's risk from three ingredients:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

What does the β ingredient do?

- A) It carries yesterday's risk into today — the memory that makes calm or stormy periods last.
- B) It reacts to the sign of yesterday's return.
- C) It sets the average return.
- D) It removes the trend from prices.

Q37. Two stocks had a volatility spike in the same crisis week. For Stock A, $\alpha + \beta = 0.95$; for Stock B, $\alpha + \beta = 0.99$. Which statement is correct?

- A) Both spikes fade at the same speed.
- B) A's spike lingers longer, because 0.95 is further from 1.
- C) Neither spike ever fades.
- D) B's storm lingers much longer: the closer $\alpha + \beta$ is to 1, the slower risk returns to normal (about three weeks to fade halfway for 0.95, over two months for 0.99).

Q38. Right after a crisis, a GARCH model shows very high volatility. What does it forecast for volatility over the coming weeks?

- A) It stays at the crisis level forever.
- B) It gradually falls back toward its long-run level.
- C) It drops to zero tomorrow.
- D) It keeps rising without limit.

Q39. A bank reports a 99% one-day Value-at-Risk (VaR) of 2% for its portfolio. What does this mean?

- A) The portfolio can never lose more than 2% in a day.
- B) The portfolio will lose exactly 2% on one day in every hundred.
- C) On about 99 days out of 100, the daily loss should be smaller than 2%; it is a threshold, not the worst case.

- D) The average daily loss is 2%.

Q40. A VaR based on one fixed, long-run volatility is broken many times, and almost all of those breaks happen during a single crisis month. What is the most likely reason?

- A) The portfolio was too small.
- B) The VaR should have used the arithmetic mean return.
- C) Risk changes over time: the fixed VaR is too tight in storms, while a GARCH VaR widens when volatility rises.
- D) Breaks of VaR can only happen in crises, so nothing is wrong.